SQL Data Analysis & Database Operations Lab

This project demonstrates practical experience in using SQL for data analysis, database management, and supporting cybersecurity-related tasks.

Key Activities

- Designed and managed relational database tables
- Performed data filtering using WHERE, AND, OR conditions
- Executed CRUD operations (Create, Read, Update, Delete)
- Used aggregate functions such as COUNT for analysis
- Applied JOIN operations to combine multiple datasets

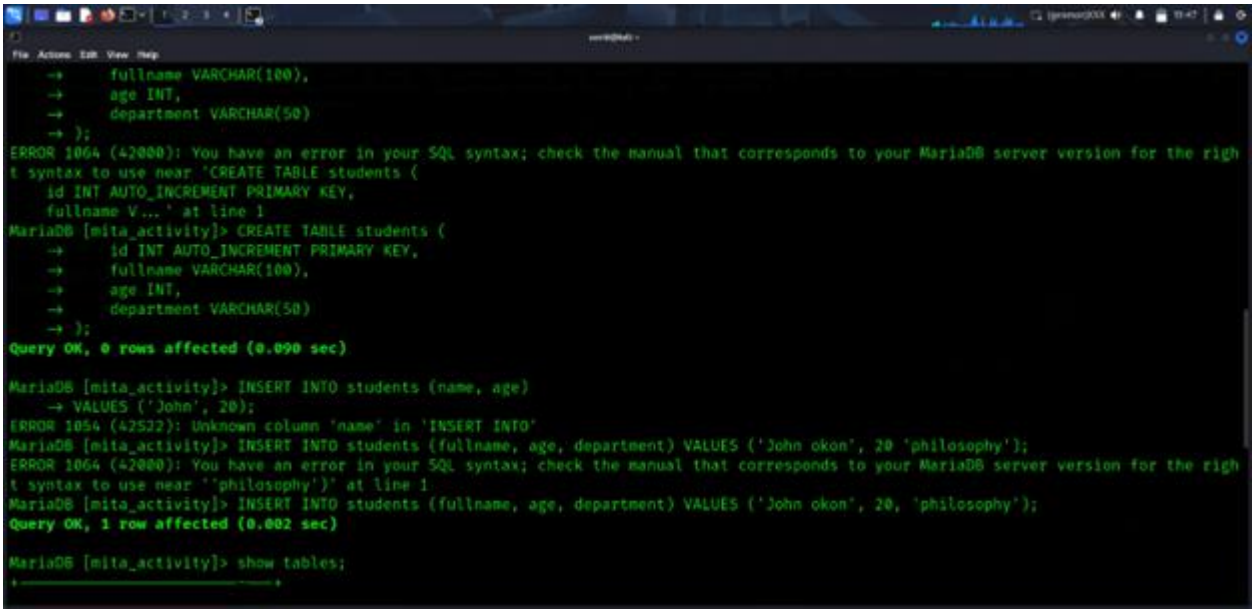
Key Skills Demonstrated

- Data querying and manipulation
- Database structuring and optimization
- Analytical thinking using SQL

Tools Used

- SQL (MySQL / SQLite)

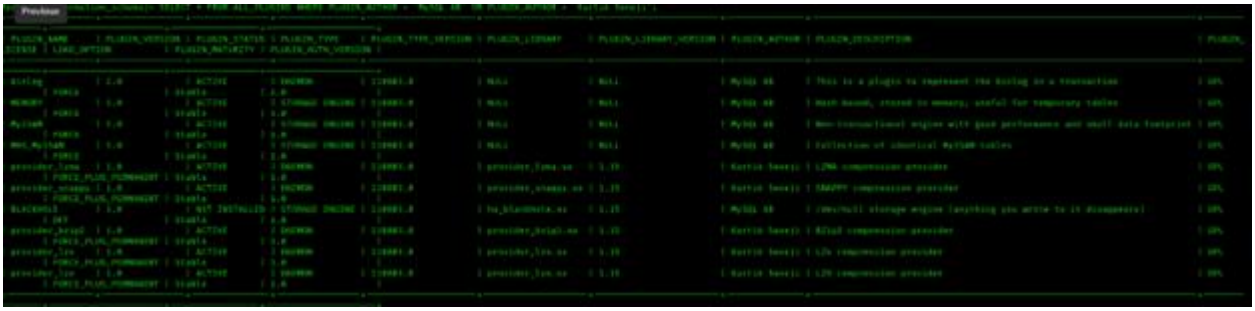
Sample Queries & Outputs



```
    ->     fullname VARCHAR(100),
    ->     age INT,
    ->     department VARCHAR(50)
    -> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the righ
t syntax to use near 'CREATE TABLE students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    fullname V...' at line 1
MariaDB [mita_activity]> CREATE TABLE students (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     fullname VARCHAR(100),
    ->     age INT,
    ->     department VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.090 sec)

MariaDB [mita_activity]> INSERT INTO students (name, age)
    -> VALUES ('John', 20);
ERROR 1054 (42S22): Unknown column 'name' in 'INSERT INTO'
MariaDB [mita_activity]> INSERT INTO students (fullname, age, department) VALUES ('John okon', 20 'philosophy');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the righ
t syntax to use near ''philosophy')' at line 1
MariaDB [mita_activity]> INSERT INTO students (fullname, age, department) VALUES ('John okon', 20, 'philosophy');
Query OK, 1 row affected (0.002 sec)

MariaDB [mita_activity]> show tables;
```

```
| STAFF_LOG                |
| student_logins           |
| students                 |
+--------------------------+
4 rows in set (0.000 sec)

MariaDB [mita_activity]> select * from students;
+----+-------------+-----+-------------+
| id | fullname    | age | department  |
+----+-------------+-----+-------------+
|  1 | John okon   |  20 | philosophy  |
+----+-------------+-----+-------------+
1 row in set (0.000 sec)

MariaDB [mita_activity]> INSERT INTO students (fullname, age, department) VALUES ('Solomon valentine', 25, 'English');
Query OK, 1 row affected (0.002 sec)

MariaDB [mita_activity]> select * from students;
+----+-------------------+-----+-------------+
| id | fullname          | age | department  |
+----+-------------------+-----+-------------+
|  1 | John okon         |  20 | philosophy  |
|  2 | Solomon valentine |  25 | English     |
+----+-------------------+-----+-------------+
2 rows in set (0.001 sec)

MariaDB [mita_activity]>
```

1. This screenshot shows the creation of a relational table used to store user data.



2. Data Filtering Using WHERE, **FROM, AND, OR** Clause
   This demonstrates filtering records based on specific conditions.

```
MariaDB [mita_activity]> UPDATE MITA_CYBERSECURITY_STUDENT SET email = 'solo99@gmaail.com' WHERE student_id = 1;
Query OK, 1 row affected (0.003 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [mita_activity]> select * from MITA_CYBERSECURITY_STUDENT;
+------------+-----------------+-----------------+-----------------+-----------+-----------------------+
| student_id | name            | department      | ip_address      | username  | email                 |
+------------+-----------------+-----------------+-----------------+-----------+-----------------------+
|          1 | SOLOMON UBONG   | ETHICAL HACKING | 192.168.100.1   | SAMBITEK  | solo99@gmaail.com     |
|          2 | SOLOMON UBONG   | ETHICAL HACKING | 192.168.100.1   | SAMBITEK  | ubsolo33@gmail.com    |
+------------+-----------------+-----------------+-----------------+-----------+-----------------------+
2 rows in set (0.000 sec)
```

3. Updating Records in a Table
   This operation demonstrates how existing records in a table were modified to ensure data accuracy and consistency by updating specific fields based on defined conditions.

```
MariaDB [mita_activity]> DELETE FROM student_logins WHERE student_id = 9;
Query OK, 1 row affected (0.003 sec)

MariaDB [mita_activity]> SELECT *FROM student_logins;
+------------+-------------+---------+-----------+-----+
| student_id | first_name  | surname | username  | age |
+------------+-------------+---------+-----------+-----+
|          1 | christopher | Okon    | carly     |  33 |
|          2 | solomon     | kali    | solo      |  51 |
|          3 | kuseme      | anthony | kuse      |  65 |
|          4 | valentine   | mita    | val       |  55 |
|          5 | joseph      | effiong | effi      |  62 |
|          6 | godwin      | alfa    | sweetbug  |  70 |
|          7 | angela      | agirl   | baby      |  40 |
|          8 | ennoch      | Stanley | phoneboy  |  44 |
+------------+-------------+---------+-----------+-----+
8 rows in set (0.000 sec)
```

4. Deleting Records from a Table
   This operation illustrates how unwanted or obsolete records were removed from the table to maintain data integrity and ensure that only relevant data is retained.

```
MariaDB [information_schema]> SELECT COUNT(PLUGIN_TYPE_VERSION) FROM ALL_PLUGINS
    -> WHERE PLUGIN_VERSION = 2.0;
+-----------------------------+
| COUNT(PLUGIN_TYPE_VERSION) |
+-----------------------------+
|                           0 |
+-----------------------------+
1 row in set (0.006 sec)
```

5. Aggregating Data Using COUNT Function
   This operation demonstrates how the COUNT function was used to determine the total number of records in a column, supporting data analysis and decision-making.

```
Previous
MariaDB [information_schema]> SELECT * FROM ALL_PLUGINS WHERE PLUGIN_AUTHOR = 'MySQL AB' OR PLUGIN_AUTHOR = 'Kartik Soneji';
+-------------+----------------+---------------+-------------+-------------------+-----------------+-----------------------+---------------+-----------------------------------------------------------------------------+---------+
| PLUGIN_NAME | PLUGIN_VERSION | PLUGIN_STATUS | PLUGIN_TYPE | PLUGIN_TYPE_VERSION | PLUGIN_LIBRARY | PLUGIN_LIBRARY_VERSION | PLUGIN_AUTHOR | PLUGIN_DESCRIPTION                                                          | PLUGIN_
LICENSE | LOAD_OPTION | PLUGIN_MATURITY | PLUGIN_AUTH_VERSION |
+-------------+----------------+---------------+-------------+-------------------+-----------------+-----------------------+---------------+-----------------------------------------------------------------------------+---------+
| binlog      | 2.0            | ACTIVE        | DAEMON      | 110803.0          | NULL            | NULL                  | MySQL AB      | This is a plugin to represent the binlog in a transaction                   | GPL
     | FORCE       | Stable        | 2.0                 |
| MEMORY      | 1.0            | ACTIVE        | STORAGE ENGINE | 110803.0          | NULL            | NULL                  | MySQL AB      | Hash based, stored in memory, useful for temporary tables                   | GPL
     | FORCE       | Stable        | 1.0                 |
| MyISAM      | 1.0            | ACTIVE        | STORAGE ENGINE | 110803.0          | NULL            | NULL                  | MySQL AB      | Non-transactional engine with good performance and small data footprint     | GPL
     | FORCE       | Stable        | 1.0                 |
| MRG_MyISAM  | 1.0            | ACTIVE        | STORAGE ENGINE | 110803.0          | NULL            | NULL                  | MySQL AB      | Collection of identical MyISAM tables                                       | GPL
     | FORCE       | Stable        | 1.0                 |
| provider_lzma | 1.0            | ACTIVE        | DAEMON      | 110803.0          | provider_lzma.so | 1.15                  | Kartik Soneji | LZMA compression provider                                                   | GPL
     | FORCE_PLUS_PERMANENT | Stable | 1.0                 |
| provider_snappy | 1.0            | ACTIVE        | DAEMON      | 110803.0          | provider_snappy.so | 1.15                  | Kartik Soneji | SNAPPY compression provider                                                 | GPL
     | FORCE_PLUS_PERMANENT | Stable | 1.0                 |
| BLACKHOLE   | 1.0            | NOT INSTALLED | STORAGE ENGINE | 110803.0          | ha_blackhole.so | 1.15                  | MySQL AB      | /dev/null storage engine (anything you write to it disappears)              | GPL
     | OFF         | Stable        | 1.0                 |
| provider_bzip2 | 1.0            | ACTIVE        | DAEMON      | 110803.0          | provider_bzip2.so | 1.15                  | Kartik Soneji | BZip2 compression provider                                                  | GPL
     | FORCE_PLUS_PERMANENT | Stable | 1.0                 |
| provider_lz4 | 1.0            | ACTIVE        | DAEMON      | 110803.0          | provider_lz4.so | 1.15                  | Kartik Soneji | LZ4 compression provider                                                    | GPL
     | FORCE_PLUS_PERMANENT | Stable | 1.0                 |
| provider_lzo | 1.0            | ACTIVE        | DAEMON      | 110803.0          | provider_lzo.so | 1.15                  | Kartik Soneji | LZO compression provider                                                    | GPL
     | FORCE_PLUS_PERMANENT | Stable | 1.0                 |
+-------------+----------------+---------------+-------------+-------------------+-----------------+-----------------------+---------------+-----------------------------------------------------------------------------+---------+
```

6. Filtering Data Using WHERE Clause
   This operation demonstrates how records were filtered based on specific conditions

```
MariaDB [mita_activity]> ALTER TABLE student_logins
    -> CHANGE COLUMN useranme username VARCHAR(255);
Query OK, 0 rows affected (0.066 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [mita_activity]> SELECT *FROM student_logins;
+------------+------------+----------+----------+-----+
| student_id | first_name | surname  | username | age |
+------------+------------+----------+----------+-----+
|          1 | christopher | Okon     | carly    |  33 |
|          2 | solomon    | kali     | solo     |  51 |
|          3 | kusene     | anthony  | kuse     |  65 |
|          4 | valentine  | mita     | val      |  55 |
|          5 | joseph     | effiong  | effi     |  62 |
|          6 | godwin     | alfa     | sweetbug |  70 |
|          7 | angela     | agirl    | baby     |  40 |
|          8 | ennoch     | Stanley  | phoneboy |  44 |
|          9 | innocent   | Bryan    | kingbee  |  24 |
+------------+------------+----------+----------+-----+
9 rows in set (0.001 sec)
```

7. Modifying Table Structure (ALTER TABLE)
   This operation illustrates how the structure of an existing table was modified, such as adding, deleting, or altering columns, to accommodate new data requirements

Conclusion

This lab demonstrates the ability to work with relational databases, perform data analysis, and support secure data handling, which are essential skills in cybersecurity operations.